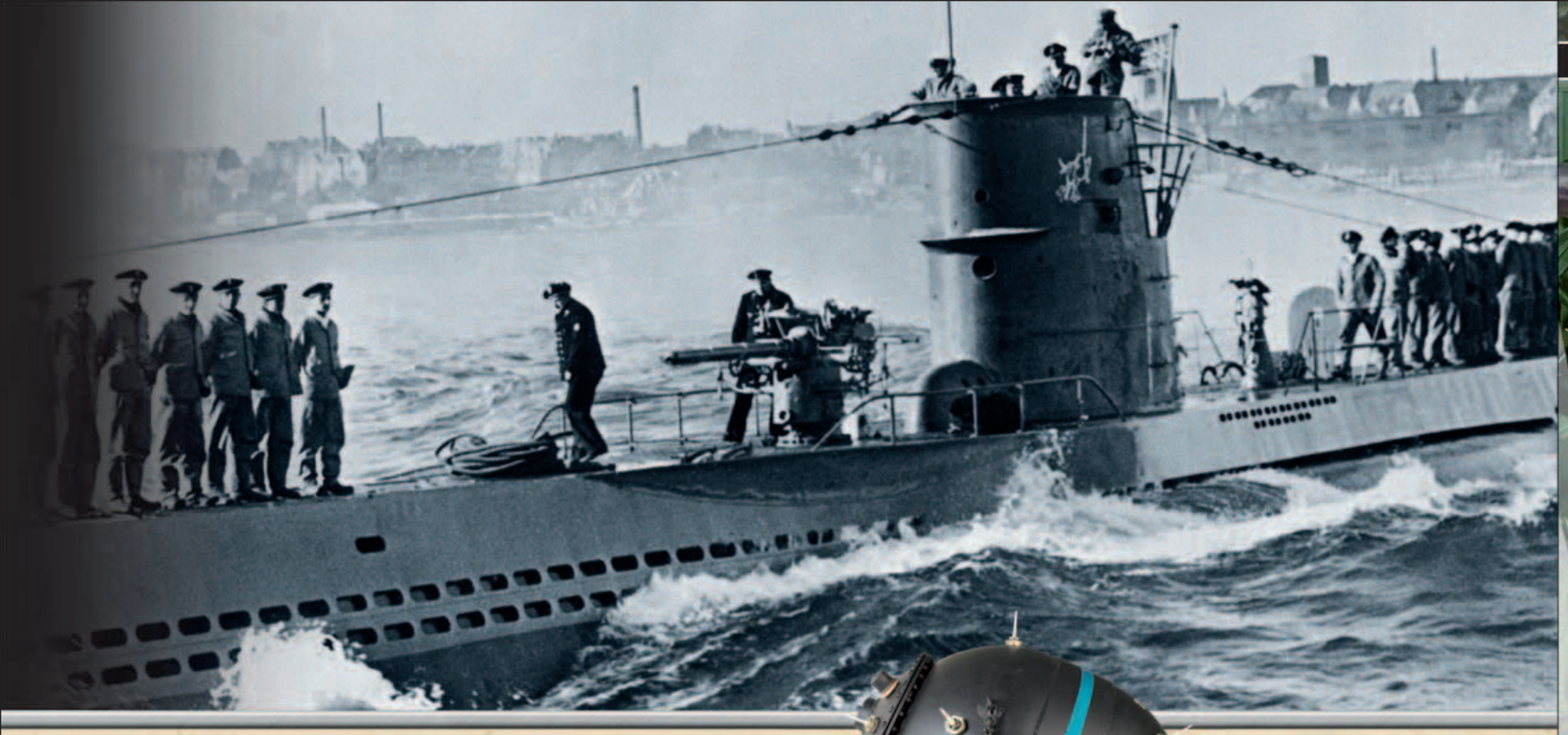


were gradually hunted down and destroyed, if at considerable cost. The Royal Navy also inflicted heavy losses on Italian warships in a number of bold actions. But the use of submarines and air power denied the British command of the sea. Although the Germans had no carriers, land-based Luftwaffe aircraft proved their deadly effectiveness against the Royal Navy in the Norwegian campaign, at Dunkirk, and in the Mediterranean, especially during the evacuation of Crete in 1941 in which 24 British warships were sunk or badly damaged by air attack. Mines were rendered less effective than in World War I through the adoption of effective countermeasures by 1940, but Axis submarines took their toll of warships as well as merchant shipping—Royal Navy ships lost to U-boat attack included the battleships *Royal Oak* and *Barham* and the carriers *Courageous* and *Ark Royal*. Convoys had to be escorted through the Mediterranean, across the Atlantic, and into the Arctic under both submarine and air attack, grueling and costly operations to keep lifelines open. It was not until 1943, more than a year after America's entry into the war, that the Allies established effective command of the Atlantic and Mediterranean. Even then much was owed to air power—the battle of the Atlantic was not won until escort carriers and long-range aircraft brought the whole ocean under cover of air patrol. The landings at Normandy in June 1944 were a staggering demonstration of naval power, but the operation depended on command of the air. The impressive productivity of Allied shipyards, especially those of the United States, meant that the odds mounted overwhelmingly against Germany as time passed. The U-boats never ceased to operate, with superior design even giving them a fresh lease of life in 1945, but it was far too little and too late.



Moored contact mine
Contact mines such as this were triggered by passing vessels brushing against the protruding horns. Other types of mine were set off by the magnetic field of a ship's hull.



Atlantic convoy
Allied survival during the war relied on a steady flow of supplies across the Atlantic. To defend merchant vessels from U-boat attack they were organized into convoys under the protection of warship escorts.